Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Instructions

1. Which of the following is the link function you should use when using the glm() function in R? — 1 / 1 point

(I) binomial, link = "logit"

(II) gaussian, link = "indentity"

(III) Poisson, link = "log"

- ○ I only
- ○ II only
- ○ III only
- ○ I and II only
- ◉ I, II, and III

✓ Correct

2. Which of the following R-command is/are related to setting up a robust regression using the MASS library? — 1 / 1 point

(I) $\mathrm{rlm}(\mathrm{y} \sim \mathrm{x}, \mathrm{data} = \mathrm{df}, \mathrm{psi} = \mathrm{psi.bisquare})$

(II) $\mathrm{rlm}(\mathrm{y} \sim \mathrm{x1} + \mathrm{x2}, \mathrm{data} = \mathrm{df}, \mathrm{psi} = \mathrm{psi.huber})$

(III) $\mathrm{glm}(\mathrm{y} \sim \mathrm{x}, \mathrm{data} = \mathrm{df}, \mathrm{family} = \mathrm{binomial})$

- ○ I only
- ○ II only
- ○ III only
- ◉ I and II only
- ○ I, II, and III

✓ Correct

3. Suppose we have the following Poisson regression model — 1 / 1 point

$$\ln(\mathrm{E}(\mathrm{Y}|\mathrm{X} = \mathrm{x})) = \ln(\lambda(x)) = 0.23 + 0.05x$$

and consider x increases its value from x_0 to $x_0 + 1$. Compute the value of

$$\frac{\lambda_1 - \lambda_0}{\lambda_0} \times 100\%$$

where $\lambda(x_0) = \lambda_0$ and $\lambda(x_0 + 1) = \lambda_1$.

- ○ 0.23%
- ○ 23%
- ○ 0.05%
- ◉ 5.127%
- ○ 25.9%

✓ Correct

4. Which of the following can be used in the variable selection process? — 1 / 1 point

(I) Ridge regression

(II) Forward stepwise regression

(III) Backward stepwise regression

- ○ I only
- ○ II only
- ○ III only
- ◉ II and III only
- ○ I, II, and III

✓ Correct

5. Let Y denote whether a student get admitted to attend Illinois Tech, with $Y = 1$ as successfully being admitted, and $Y = 0$, otherwise. Also, let — 1 / 1 point

- X_1 denote the math scores (0-100)
- X_2 denote the reading scores (0-100)

of a particular standardize exam that a student need to take to gain admission. We collected n-data and run a logistic regression to obtain

$$\ln\left(\frac{\pi}{1-\pi}\right) = 0.13 + 0.05X_1 + 0.04X_2$$

where

$$\pi = \Pr(Y = 1|X_1 = x_1, X_2 = x_2).$$

Suppose a student has a math scores of 78% and the reading scores of 82%. If the student retake the math test and scores 83%. Compute the following change of odds in being admitted to Illinois Tech.

$$\frac{\mathrm{Odds}(x_0+1) - \mathrm{Odds}(x_0)}{\mathrm{Odds}(x_0)} \times 100\%$$

- ◉ 5.127%
- ○ 13.883%
- ○ 4.081%
- ○ 0.05%
- ○ 0.04%

✓ Correct

6. Which of the following is an advantage of using the M-estimation technique? — 1 / 1 point

- ○ To detect and remediate multicollinearity issue in the data.
- ○ To obtain estimations of the regression parameters using the least square method.
- ○ To detect and delete outliers in the data.
- ○ M-estimation is used as a variance stabilizing technique.
- ◉ M-estimation is a robust regression technique used when the data has outliers.

✓ Correct

7. Which of the following can be used to measure how good a model can predict the actual observations. — 1 / 1 point

(I) A high R^2 score.

(II) A high MAE score.

(III) A high RMSE score.

- ◉ I only
- ○ II only
- ○ III only
- ○ II and III only
- ○ I, II, and III

✓ Correct

8. Which of the following statement(s) is/are true about a binary logistic regression? — 1 / 1 point

(I) The predictor variables (x) are continuous, with values between 0 and

(II) The response variable (y) is continuous, with values between 0 and 1.

(III) The outputs of the sigmoid function are continuous, with values between 0 and 1.

- ○ I only
- ○ II only
- ◉ III only
- ○ II and III only
- ○ I, II, and III

✓ Correct

9. In the k-fold cross-validation technique, the data are shuffled randomly and then divided into k equal sub-samples. Which of the following statement(s) is/are true? — 1 / 1 point

(I) We then use all the samples as training samples to be learned from.

(II) We then use $k - 1$ samples as training samples and the k-th sample as a test sample.

(III) We then use $k - 2$ samples as training samples, and 2 samples as test samples.

- ○ I only
- ◉ II only
- ○ III only
- ○ II and III only
- ○ I, II, and III

✓ Correct

10. Suppose we want to model the profitability of a movie (y) based on the opening-weekend revenue (x, in millions of dollar). We let y = 1 if the movie is profitable, that is, the revenue is larger than the budget, and not profitable as y = 0. Based on the data from 100 movies in the past 50 years, we obtain — 1 / 1 point

$$\log(\mathrm{odds}) = -2.27 + 1.276x$$

Which of the following is/are correct?

(I) The odds ratio is given by $e^{1.276}$.

(II) For a one-unit increase in x, we expect a 258.2% increase in the odds.

(III) For a one-unit increase in x, we expect the odds that the movie will be profitable increase by a factor of 3.58.

- ○ I only
- ○ II only
- ○ III only
- ○ II and III only
- ◉ I, II, and III

✓ Correct

11. Which of the following component(s) is/are involved in a generalized linear model? — 1 / 1 point

(I) An exponential family model for the response variable y.

(II) A link function that connects the means of the response to the linear predictors.

(III) Predictor variables having a normal distribution.

- ○ I only
- ○ II only
- ○ III only
- ◉ I and II only
- ○ I, II, and III

✓ Correct

12. Suppose Y is a binary response variable, taking values either 0 or 1, while X_1 and X_2 are predictor variables. Predicting the probability — 1 / 1 point

$$\Pr(Y = 1|X_1, X_2)$$

involves setting up a

- ○ Multiple linear regression model
- ○ Simple linear regression model
- ○ Poisson regression model
- ◉ Logistic regression model
- ○ Gamma regression model

✓ Correct

13. Which of the following statement(s) is/are true? — 1 / 1 point

(I) We say that a random variable Y follows the Poisson distribution if Y takes values in {0, 1, 2, . . . } and

$$\Pr(Y = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

where $k = 0, 1, 2, 3, \cdots$, for some $\lambda > 0$.

(II) If Y is a Poisson random variable, then the mean of Y is the same as the variance of Y.

(III) In Poisson regression, we assume that the mean of a Poisson random variable Y can be written as

$$E(Y|X_1, X_2, \cdots, X_n) = \beta_0 + \beta_1 X_1 + \cdots + \beta_n X_n.$$

- ○ I only
- ○ II only
- ○ III only
- ◉ I and II only
- ○ I, II, and III

✓ Correct

14. Suppose that the probability of an event is 0.2, what is the odds of the event occurring? — 1 / 1 point

- ○ 0.2
- ○ 0.8
- ○ 0.12
- ◉ 0.25
- ○ 0.3

✓ Correct